

WaveFlow-UIE

A Single-Pipeline **Frequency-Guided Flow Model** for Underwater Image Restoration

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April 2026

Agenda

What we'll cover today

01

The Problem

Underwater degradation & why it matters



02

Background

Frequency-domain insights & WF-Diff baseline



03

Our Method

WaveFlow-UIE — a single wavelet-flow pipeline



04

Experiments

Quantitative, qualitative & speed analysis



→ 17 slides · ~10 min · Q&A at the end

Underwater Images Are Degraded

And that breaks downstream computer-vision systems

Light absorption & scattering

produce **color cast**, **low contrast**, **haze**, and **lost texture detail**.

WHY IT MATTERS

- 🐟 Marine inspection & biology
- 🤖 Underwater robotics & navigation
- 👁️ Object detection & tracking

color cast →

← *haze veil*

blurred texture →

Degraded underwater scene

Baseline: WF-Diff (CVPR 2024)

Strong frequency-aware results — but a heavy two-stage pipeline

STAGE 1

WFI²-net

Coarse wavelet-domain enhancement via
Transformer + Fourier blocks



STAGE 2

FRDAM (diffusion)

Two diffusion branches refine LL & HF residuals
separately



THE COST

Two networks + iterative diffusion sampling → **~0.28s / image** inference and high architectural complexity.

Our Idea: Collapse the Pipeline

Replace coarse-then-refine with a single learned flow

WF-Diff

two-stage

Coarse Net (WFI²-net)



Diffusion Branch (LL)



Diffusion Branch (HF)

VS

WaveFlow-UIE

single pipeline



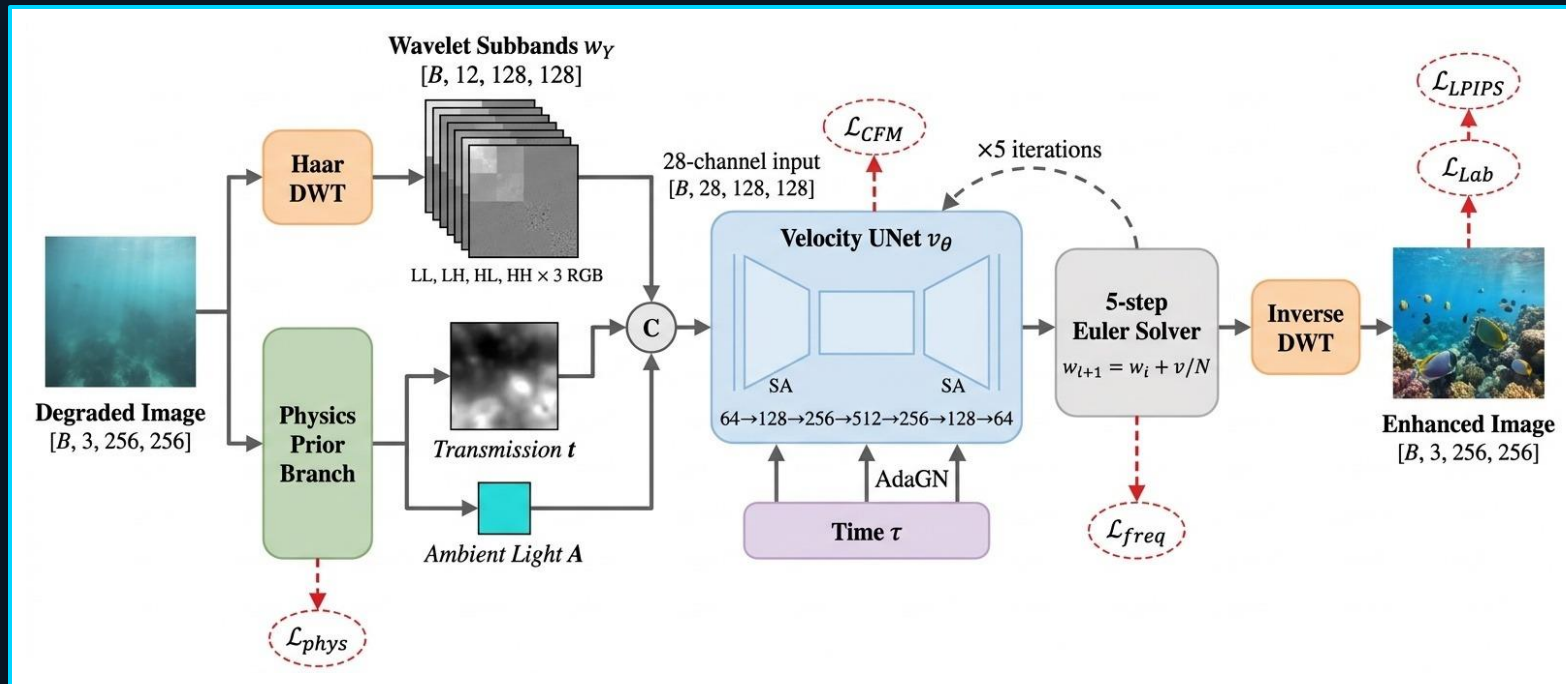
Conditional Velocity Network

learns wavelet-space flow direction

One network. Frequency-aware. Physics-conditioned. Trained with flow matching.

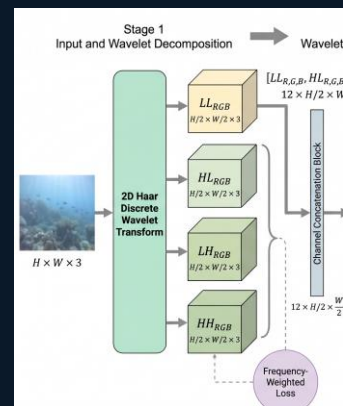
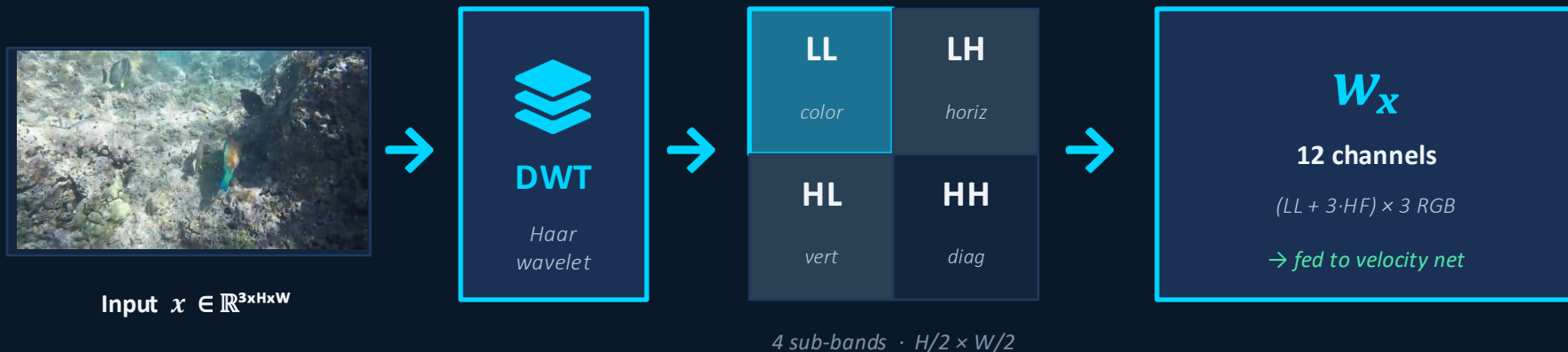
Detailed Architecture

Tensor shapes, sub-band layout, and loss attachment points



Inside the Wavelet Decomposition

Splitting the image into bands the model can reason about separately



Why this helps: degradation is **non-uniform across bands** · color sits in LL, fine detail in HF — process them with different weights.

Inside the Physics Prior

4 conv layers + 2 heads — predicts transmission t and ambient light A



Input x

$3 \text{ ch} \cdot H \times W$



Conv 3×3

$3 \rightarrow 32 \text{ ch}$
+ ReLU



Conv 3×3

$32 \rightarrow 64 \text{ ch}$
+ ReLU



Conv 3×3

$64 \rightarrow 32 \text{ ch}$
+ ReLU



BACKBONE · 3 conv layers



Conv 1×1

$32 \rightarrow 1$ + Sigmoid

t · transmission map

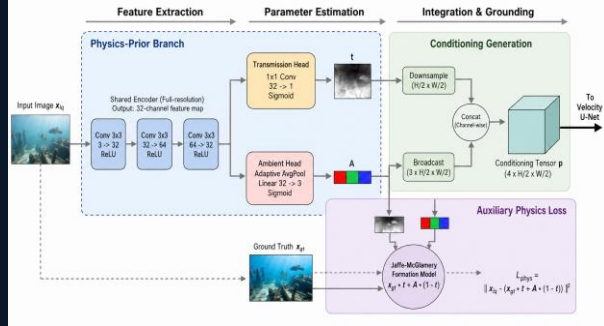
$(B, 1, H, W)$ · per-pixel

AvgPool → FC

$32 \rightarrow 3$ + Sigmoid

A · ambient light

$(B, 3) \rightarrow \text{broadcast}$

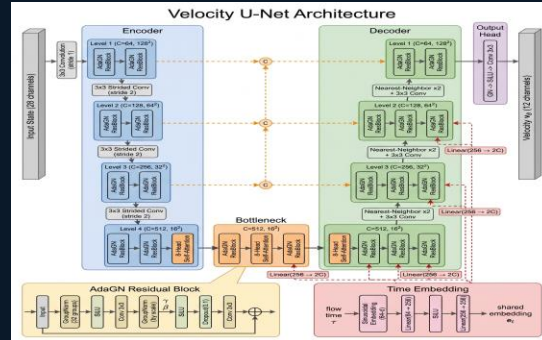


$p = [t, A] \rightarrow 4 \text{ channels}$, downsampled to $H/2 \times W/2$ · fed to velocity UNet

Underwater formation model: $x \approx y \cdot t + A \cdot (1 - t)$

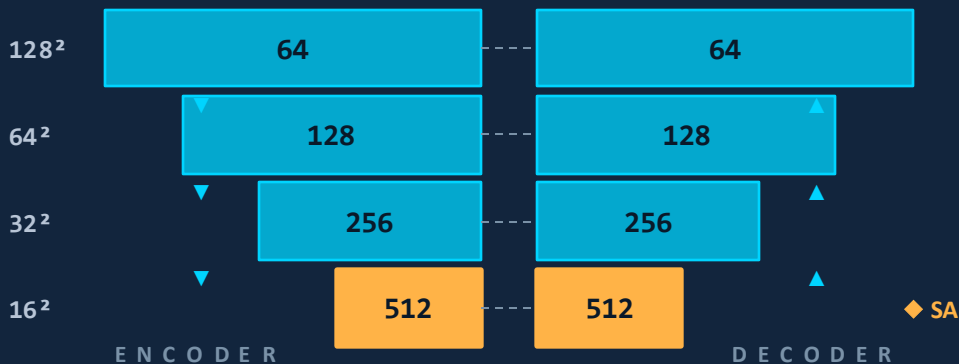
Inside the Velocity Network

AdaGN U-Net · $64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ · self-attention at 16^2



U - NET

4 levels · 3 down / 3 up



AdaGN RESBLOCK

how time τ enters every block

GroupNorm (32 groups)

SilU

Conv 3x3

AdaGN · scale & shift $\leftarrow \tau$

SilU + Dropout 0.1

Conv 3x3

$\tau \rightarrow \text{sinusoidal}(64) \rightarrow \text{MLP} \rightarrow 256-d$

◆ self-attention at 16^2 · 8 heads $\times$ 64 dim · skip = concat · down: stride-2 conv · up: nearest + conv

INPUT 28 ch ($12 w_t + 12 w_x + 4 [t, A]$) · OUTPUT 12 ch velocity $\hat{v}$ ·

Training Objective

Five complementary losses guide what the flow learns

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{cfm} + \lambda_2 \mathcal{L}_{freq} + \lambda_3 \mathcal{L}_{lpips} + \lambda_4 \mathcal{L}_{lab} + \lambda_5 \mathcal{L}_{phys}$$

$\mathcal{L}_{cfm}$

Flow Matching

predict the right velocity
in wavelet space

$\mathcal{L}_{freq}$

Frequency

weighted L1 — extra
weight on high-freq detail

$\mathcal{L}_{lpips}$

Perceptual

deep-feature similarity
(LPIPS)

$\mathcal{L}_{lab}$

Color (Lab)

consistent color in
perceptual color space

$\mathcal{L}_{phys}$

Physics

$x \approx y \cdot t + A \cdot (1 - t)$

Frequency loss carries the largest weight on high-frequency components — preserves edges & texture

Datasets & Setup

Trained and evaluated across four standard underwater benchmarks

UIEB**890**

images

real-world · paired

800 train · 90 test

LSUI**4279**

images

diverse scenes · paired

3879 train · 400 test

UFO-120**120**

images

USR-248-style · paired

test set

EUVP**515**

images

Images · paired

test set

EVALUATION SETUP

RESOLUTION

256×256 & native

METRICS

FID · LPIPS · PSNR · SSIM

NON-REFERENCE

UIQM · UCIQE on U45 / C60

HARDWARE

single GPU · consistent runtime

Hyperparameters & Setup

Reproducibility — what we trained, how, and where

Training

Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)
Learning rate	1×10^{-4}
Batch size	8
Resolution	256×256
Epochs	200
Scheduler	Cosine decay

Loss Weights

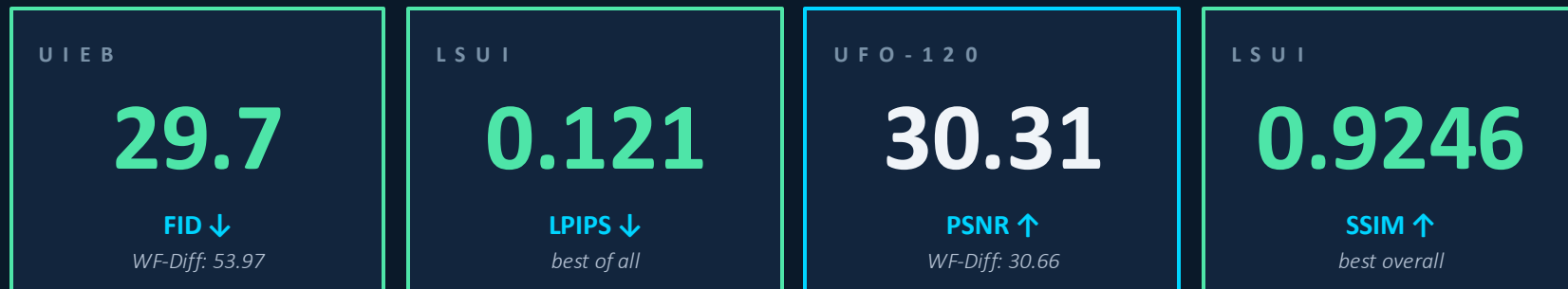
λ_{cfm}	1.0	<i>flow matching</i>
λ_{freq}	0.5	<i>frequency L1</i>
λ_{lpips}	0.1	<i>perceptual</i>
λ_{lab}	0.2	<i>Lab color</i>
λ_{phys}	0.05	<i>physics consistency</i>
α (HF)	2.0	<i>high-freq weight</i>

Inference + HW

Flow steps K	5
Step size Δt	1 / K
Inference time	~79 ms (256^2)
GPU	RTX 5080 (16GB)
Framework	PyTorch 2.x
Wavelet	Haar

Quantitative Results

Best-or-second-best across UIEB, LSUI, UFO-120 and EUVP



vs. SOTA baselines (256×256, lower-is-better ↓ / higher-is-better ↑)

	WF-Diff	SCNet	U-shape	Ours
UIEB FID ↓	53.97	30.90	48.63	29.74
LSUI LPIPS ↓	0.151	0.206	0.145	0.122
UFO-120 PSNR ↑	30.66	23.48	29.55	30.31
EUVP FID ↓	23.09	29.76	22.95	26.36

Ablation Study

What does each component actually contribute?

Configuration	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	UCIQUE ↑	UIQM ↑
Full model (Ours)	<u>24.9818</u>	<u>0.9316</u>	<u>0.1390</u>	<u>29.7382</u>	0.5835	4.0450
No physics prior branch	24.5000	0.9234	0.1469	35.7000	<u>0.5853</u>	4.0912
No frequency loss weighting	24.0902	0.9298	0.1426	33.8364	0.5852	4.0270
No LPIPS loss	24.0288	0.9047	0.1574	35.1146	0.5753	<u>4.2561</u>
No lab color loss	24.1482	0.9220	0.1848	44.0632	0.5819	4.0640
K = 1 (single step)	24.9382	0.9306	0.1403	30.6582	0.5827	4.0638



WHAT WE EXPECT TO SEE

Removing the **physics prior** hurts color accuracy. Removing the **frequency loss** softens fine detail. **K=1** trades quality for speed.

Qualitative Results

Restored color and detail across diverse underwater scenes

Input



WF-Diff



WaveFlow-UIE

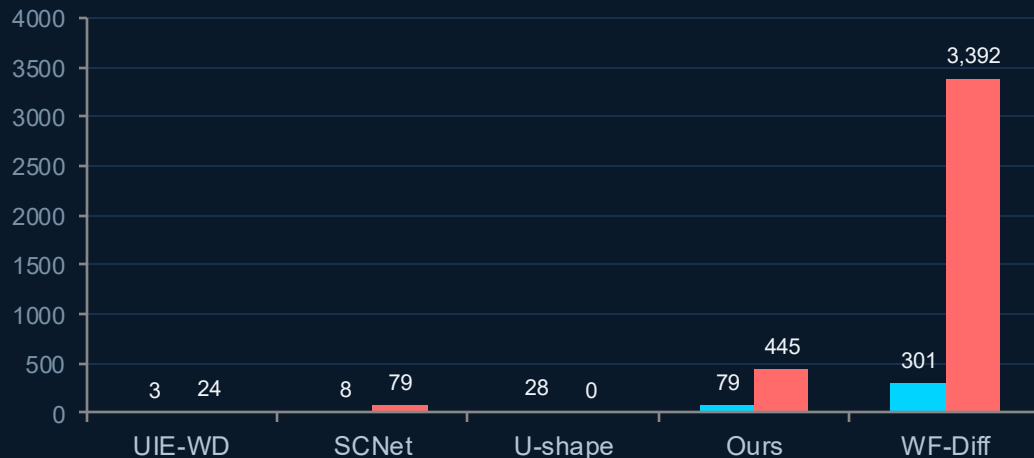


Speed & Tradeoffs

Faster than diffusion-based WF-Diff — honest about where we sit

Inference time · ms / image ↓ (UIEB, log scale)

■ 256 × 256 ■ Native resolution



3.8× / 7.6× faster

at 256² / native res · vs WF-Diff

- ✓ Single-pipeline simplicity
- ✓ Best FID on UIEB / UFO-120
- ✓ Best LPIPS & SSIM on LSUI
- ⚖ Slower than purely-feedforward CNNs (still iterative)
- ⚖ Native-resolution cost grows with image size

Takeaways & Future Work

What we showed — and where it goes next

CONTRIBUTIONS



Reformulated WF-Diff as a single wavelet-domain flow



Added a physics prior branch for transmission & ambient



Frequency-aware losses: detail + color jointly preserved



Faster inference at competitive — often better — quality

FUTURE WORK



Reduce flow steps (1-step distillation)



Test on real-time underwater video streams



Evaluate on downstream detection / SLAM



Stronger physics priors (depth-aware)

Thank you.